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Lattice-Boltzmann simulation of Two-phase flow in carbonate porous media retrieved from computed Microtomography

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HIGHLIGHTS

- A natural multimodal carbonate rock is imaged and processed by a high- resolution micro-CT to extract a 3-D geometry model.
- A multiple-relaxation-time color-gradient LB model is developed to simulate two-phase flow in carbonate porous media.
- The effects of rock wettability, capillary number and viscosity ratio on displacement efficiency are explored.
- The underlying mechanisms of pore-scale oil droplets' mobilization are understood by integral geometry.

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ABSTRACT

Marine carbonate reservoirs widely distributed in the Middle East are characterized by multimodal pore structures and complex wettability, usually causing a great difference in pore-scale oil displacement efficiency. However, the underlying mechanisms are still ambiguous. In this study, a natural rock sample selected from a typical carbonate reservoir in the Middle East is imaged with a high-resolution X-ray micro-CT, and the three-dimensional geometry model of microstructures are extracted. A multiple-relaxation-time color-gradient lattice Boltzmann model is developed and validated to simulate oil–water two-phase flow in the carbonate porous media. The influences of rock wettability, oil-wet heterogeneity, capillary number, and oil–water viscosity ratio on oil displacement efficiency and fluid distribution in multimodal carbonate pore space are ultimately explored, following by analysis of pore-scale oil droplets mobilization with integral geometry. Results show that, the rock wettability, capillary number and oil–water viscosity ratio have significant impacts on the pore-scale oil displacement efficiency and fluid distribution in multimodal carbonate pore space while oil-wet heterogeneity has little effect. The oil displacement efficiency usually becomes larger with the increase of capillary number and the water-wetting degree as well as the decrease of oil–water viscosity ratio. Due to dynamic competition between capillary pressure and viscous force, the continuous oil droplets are fragmented into a large quantity of isolated oil droplets in the early stage of water flooding, showing a sharp decrease in the volume of continuous oil droplets, a rapid increase in the volume of isolated oil droplets, and a poor topological connectivity; In the middle and late stage of water flooding, the isolated oil droplets are gradually stripped and mobilized, leading to a decrease in the volume of oil droplets and an obvious improvement in topological connectivity.

1. Introduction

According to the latest BP energy report (Statistical review of world energy.N.D.), the remaining recoverable oil reserve in the Middle East is about 1132 $\times$ 10^8t at 2021, accounting for 48.3 % of the world's remaining recoverable oil reserve, and Middle East crude oil production accounts for 31.3 % of global crude oil

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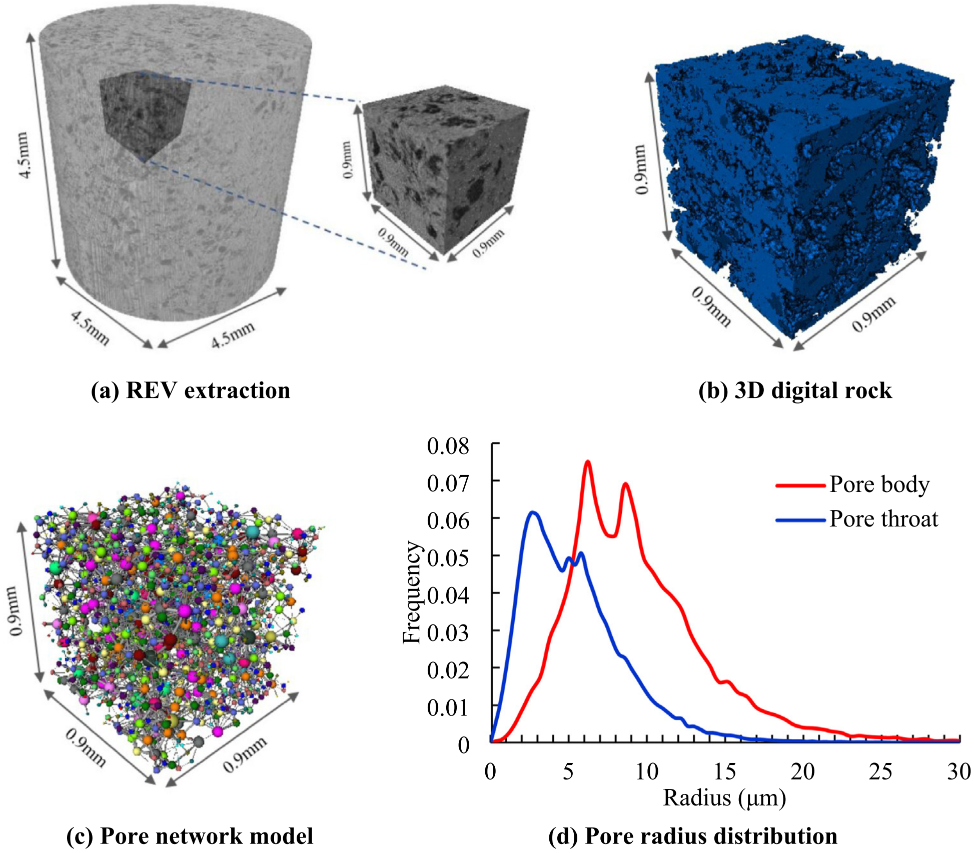


Fig. 1. Workflow for micro-CT image processing and analysis of a multimodal carbonate rock.

production. In last decades, China's oil and gas business has expanded rapidly in the Middle East, where the most important production base overseas has been established. There exist abundant marine carbonate reservoirs in the Middle East dominated by bioclastic limestone, which are quite different from the naturally fracture-vuggy carbonate reservoirs in China (Wang et al., 2016; Wang et al., 2020; Li et al., 2020; Ning et al., 2021; Zhang et al., 2022). Due to strong reservoir heterogeneity, the widely distributed barriers and thief zones resulted in a low sweep efficiency and a poor producing degree of recoverable reserve during water flooding. Efficient exploitation of the remaining recoverable oil reserve in carbonate reservoirs of the Middle East faces great challenges (Li et al., 2020; Li et al., 2021; Sun et al., 2022). Song et al (Song and Li, 2018) suggested that the marine carbonate reservoirs found in the Middle East are occupied by multiple types of pores with different origin and size, including bio-framework pores, intergranular dissolved pores and matrix micro-pores. They are commonly characterized by multimodal pore structure, causing a poor correlation between porosity and permeability and the difference in permeability under the same porosity is higher than 10,000 times. Moreover, the rock wettability is extremely complicated, mainly mixed-wet or oil-wet, which further aggravates the difference of waterflooding production performance. Therefore, it is of great importance to further improve oil recovery by investigating the oil–water two-phase flow in multimodal carbonate porous media to clarify the underlying mechanism of pore-scale oil displacement efficiency.

The advance of digital rock physics has provided an efficient alternative to investigate the multi-phase fluid flow in porous media. The commonly used methods for building digital rock include physical experiments and numerical reconstruction (Blunt et al., 2013). As a non-destructive technique with a submicron voxel resolution, the X-ray micro-CT is not only adapted to evaluate the microstructure of various rock types but also monitor the fluid flow process by combining with an indoor coreflooding system. Iglauer et al (Iglauer et al., 2012) originally performed CT imaging of water-wet and oil-wet sandstones to explore the differences in size, morphology and quantity of cluster-like remaining oil droplets during the late stage of waterflooding. Lin et al (Lin et al., 2016) and Bijeljic et al (Bijeljic et al., 2018) investigated the influence of micro-pore distribution in carbonate rock on single-phase flow based on X-ray micro-CT differential imaging, and estimated the connectivity between macro-pores. Gao et al (Gao et al., 2019) subdivided the pores of water-wet carbonate rock into macro-pores and micro-pores, analyzed the steady-state flow of oil and water using micro-CT imaging, and clarified the significant contribution of movable oil in micro-pores to waterflood displacement efficiency and its influence on steady-state relative permeability. Liu et al (Liu et al., 2017) integrated CT imaging and integral geometry to observe the microstructure of two-phase fluids and its control mechanism on relative permeability. It demonstrated that the fluid phase topology and connectivity at pore scale have an important impact on relative permeability. Reynolds et al (Reynolds et al., 2017) further conducted X-ray CT imaging to investigate the impact of low capillary number on the interpore connectivity in Bentheimer sandstone. Zou et al (Zou et al., 2018) further evaluated the relationship between relative permeability, fluid distribution and topological connectivity under water-wet and mixed-wet conditions. Results demonstrated that under mixed-wet condition, increased dynamic connectivity and ganglion dynamics will result in a non-equilibrium effect at the fluid–fluid interface, causing more energy dissipation during fractional flow in mixed-wet conditions and a lower effective permeability than that obtained under water-wet conditions. Spurin et al (Spurin et al., 2019) confirmed the intermittent flow between micropores and macropores for a carbonate rock sample using microfocus CT imaging. The competition will move to smaller pores when the displacement rate is increased. The intermittent flow typically induces the occurrence of fluid snap-off at pore-scale which exerts an important impact on fluid relative permeability. Through 3D micro-CT imaging experiments, Zhang et al (Zhang et al., 2022) demonstrated that when the fluid flow in porous media is mainly driven by capillary pressure, there is a nonlinear intermittent flow regime with a power-law dependence between pressure gradient and flow rate. As introduced above, the oil–water two-phase flow in multimodal carbonate rock is more complicated than that in sandstone with relatively uniform pore structure. Many efforts have been made to discover the existence of intermittent porethis work, the Navier-Stokes equations are used to describe oil–water two-phase flow in porous media, as shown in equation (1).

$$\nabla \cdot \boldsymbol{u} = 0 \tag{1}$$

$$\partial_t(\rho\boldsymbol{u}) + \nabla \cdot (\rho\boldsymbol{u}\boldsymbol{u}) = -\nabla \cdot p + \nabla \cdot \left[\eta\left(\nabla\boldsymbol{u} + \nabla\boldsymbol{u}^T\right)\right] + \boldsymbol{F}_s + \boldsymbol{G} \tag{2}$$

Where$\boldsymbol{u}$,t,ρ, η and p are the velocity vector, time, fluid density, dynamic viscosity and pressure, respectively; $\boldsymbol{G}$ is the body force; $\boldsymbol{F}_s$ is the interfacial force.

Instead of using the traditional computational fluid dynamics (CFD) methods (Iglauer et al., 2012) (such as finite difference method, finite element method, volume of fluid method, level-set method and phase-field method, etc.) to solve the Navier-Stokes equations with appropriate numerical schemes, the LBM has unique advantages in dealing with fluid–fluid interface interaction and boundary conditions. LBM originated from the discrete schemes of Boltzmann dynamic equations. During LBM modeling, different fluids are simplified as fictitious particles characterized by particles with distribution functions, and the particles are subjected to consecutive propagation and collision over a discrete lattice grid. The velocity and density of one fluid are calculated from its distribution function. The Boltzmann equation reflects the space–time evolution of particle distribution function, which is expressed as:

$$\frac{\partial f}{\partial t} + \boldsymbol{e}\frac{\partial f}{\partial \boldsymbol{x}} + a\frac{\partial f}{\partial \boldsymbol{c}} = \left(\frac{\partial f}{\partial t}\right)_{collision} \tag{3}$$

where f denotes to the particle distribution function, which is a function of time, position and velocity; $(\partial f/\partial t)_{\mathrm{collision}}$ is a collision term described by a complex nonlinear integral term. To solve this problem, Bhatnagar, Gross and Krook proposed a single-relaxation-time collision operator (Fu et al., 2016). This operator can satisfy the conservation of mass, momentum, and energy. It can effectively describe the trend of fluid system towards equilibrium, and simplify the particle collision as a “relaxation” process towards the equilibrium particle distribution function. The single-relaxation-time collision operator is described by

$$\left(\frac{\partial f}{\partial t}\right)_{collision} = \frac{f_i(x,t) - f_i^{eq}(x,t)}{\tau} \tag{4}$$

where τ is the dimensionless relaxation time, representing the average time interval between two particle collisions, also called as the collision time; $f_i^{eq}(\boldsymbol{x},t)$ is the equilibrium particle distribution function.

The above BGK Boltzmann equation is further discretized with the single-relaxation-time collision operator, which takes the form of:

$$f_i(x + c_i\Delta t, t + \Delta t) - f(x,t) = -\frac{f_i(x,t) - f_i^{eq}(x,t)}{\tau} \tag{5}$$

In the lattice Boltzmann method, a function of particle distribution f_i is used to solve the dynamic evolution of fluid, which is dependent on the discrete velocity$\boldsymbol{c}_i$. The DnQb discrete velocity models proposed by Qian et al were widely used to simulate the two-dimensional space (D2Q9) or the three-dimensional space (D3Q13, D3Q15, D3Q19, D3Q27) (Bijeljic et al., 2018; Gao et al., 2019; Liu et al., 2017; Reynolds et al., 2017), as shown in Fig. 2, where D represents the spatial dimension and Q represents the discrete velocity direction. In the D3Q19 model, the discrete velocity $\boldsymbol{c}_i$ is:

$$\boldsymbol{c}_i = \begin{cases} \{0,0,0\}^T, i = 0 \\ \{\pm1,0,0\}^T, i = 1,2 \\ \{0,\pm1,0\}^T, i = 3,4 \\ \{0,0,\pm1\}^T, i = 5,6 \\ \{\pm1,\pm1,0\}^T, i = 7,8,9,10 \\ \{\pm1,0,\pm1\}^T, i = 11,12,13,14 \\ \{0,\pm1,\pm1\}^T, i = 15,16,17,18 \end{cases} \tag{6}$$

The equilibrium distribution function in DnQb model can be generalized as:

$$f_i^{eq} = \omega_i\rho\left[1 + \frac{1}{c_s^2}(\boldsymbol{c}_i \cdot \boldsymbol{u}) + \frac{1}{c_s^4}(\boldsymbol{c}_i \cdot \boldsymbol{u})^2 - \frac{1}{2c_s^2}\boldsymbol{u}^2\right] \tag{7}$$

where ρ is the fluid density; ω_i is the weight coefficient. For D2Q9 model, ω_0 = 4/9, $\omega_{1\text{-}4}$ = 1/9, $\omega_{5\text{-}8}$ = 1/36; For D3Q19 model, ω_0 = 1/3, $\omega_{1\text{-}6}$ = 1/18, $\omega_{7\text{-}18}$ = 1/36; $\boldsymbol{u}$ is the particle velocity vector on each grid point; c_s is the pseudo speed of sound, for D2Q9 and D3Q19 models,$c_s = 1/\sqrt{3}$.

Previous studies (Mehmani et al., 2020; Liu et al., 2021; Li et al., 2015) have reported that the stability of lattice BGK model strongly depends on the single-relaxation-time, and the reliability of simulated results is poor when the viscosity ratio is high. To overcome this issue, a relaxation matrix is introduced to replace the single-relaxation-time, and a multiple-relaxation-time collision operator is developed. The stability and accuracy are effectively improved by adjusting multiple free parameters.

The multiple-relaxation-time collision operator Ω_i can be expressed as:

$$\Omega_i(\boldsymbol{x},t) = -\boldsymbol{M}^{-1}\boldsymbol{S}[\boldsymbol{m}(\boldsymbol{x},t) - \boldsymbol{m}^{eq}(\boldsymbol{x},t)]\Delta t \tag{8}$$

where $\boldsymbol{S}$ is a diagonal relaxation matrix,$\boldsymbol{S} = \omega\boldsymbol{I} = diag(\omega,\cdots\omega)$; $\boldsymbol{M}$ is the transformation matrix from discrete velocity space to momentum space,$\boldsymbol{m} = \boldsymbol{M}f$; $\boldsymbol{m}^{eq}$ is the equilibrium vector in moment space,$\boldsymbol{m}^{eq} = \boldsymbol{M}f^{eq}$; $\boldsymbol{M}^{-1}$ is the inverse of the transformation matrix$\boldsymbol{M}$.

Based on the treatment of fluid–fluid interaction, multi-phase multi-component LB models (Fu et al., 2016; Wei et al., 2018; Sharma et al., 2019). are further subdivided into color-gradient model, Shan-Chen model, pseudo-potential model and free energy model The color-gradient model and pseudo-potential model are most widely used. The former is suitable for simulating immiscible two-phase flow and relatively thin fluid interface, with a wide range of viscosity ratios. The latter is applicable to gas–liquid two-phase flow with a relatively thick fluid interface and a small range of viscosity ratios where the interfacial tension is strongly related to fluid viscosity. Previous studies (Li et al., 2019; Ning et al., 2019) showed that compared with other LB models, the color

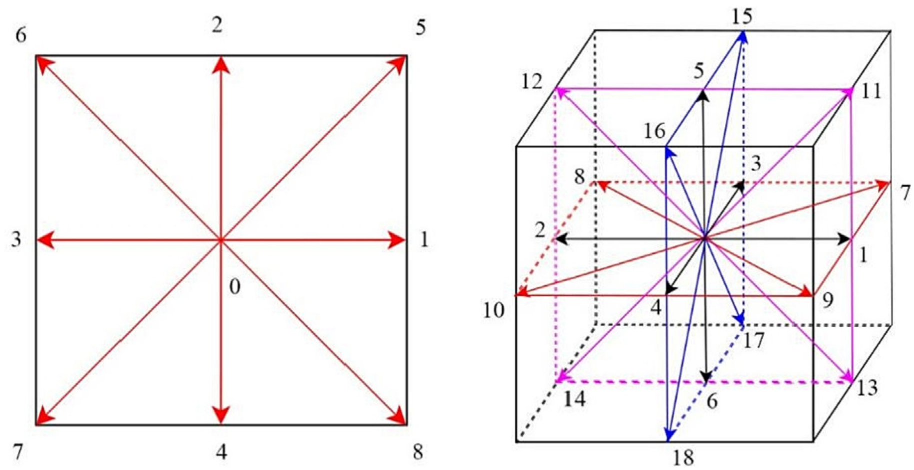


Fig. 2. Discrete velocity model (left: D2Q9, right: D3Q19).

gradient model is more efficient to simulate the wettability change. Therefore, the color gradient model is selected to simulate two-phase flow in multimodal carbonate porous media, and explore the influences of wettability, oil-wet heterogeneity, capillary number and viscosity ratio on oil displacement efficiency.

In the color-gradient LB model (Mehmani et al., 2020), two distribution functions ($f_i^{\mathrm{R}}(\boldsymbol{x},t)$ and $f_i^{\mathrm{B}}(\boldsymbol{x},t)$) are introduced to represent two immiscible fluids (red fluid and blue fluid). The subscript i denotes to the velocity direction of the i lattice. The total distribution function is defined as $f_i(\boldsymbol{x},t) = f_i^{\mathrm{R}}(\boldsymbol{x},t) + f_i^{\mathrm{B}}(\boldsymbol{x},t)$, and the total fluid density is $\rho(x,t) = \rho_R(x,t) + \rho_B(x,t)$. Each simulation includes collision step, recolor step and propagation step.

After the collision step, the total distribution function is described as:

$$\tilde{f}_i(\boldsymbol{x},t) = f_i(\boldsymbol{x},t) + \Omega_i(\boldsymbol{x},t) + F_i \tag{9}$$

where $\tilde{f}_i(\boldsymbol{x},t)$ is the distribution function after collision; F_i is the external force term, which can be expressed as follows:

$$F_i = \sum_j \left(\boldsymbol{I} - \frac{1}{2}\boldsymbol{M}^{-1}\boldsymbol{SM}\right)_{ij} \omega_j \left[\frac{\boldsymbol{c}_j - \boldsymbol{u}}{c_s^2} + \frac{(\boldsymbol{c}_j \cdot \boldsymbol{u})\boldsymbol{c}_j}{c_s^4}\right] \cdot (\boldsymbol{F}_s + \boldsymbol{G})\Delta t \tag{10}$$

$$\boldsymbol{F}_s = \frac{1}{2}\left|\nabla \rho^N\right| (\sigma \kappa \boldsymbol{n} + \nabla_s \sigma) \tag{11}$$

where the phase field $\rho^{\mathrm{N}} = (\rho^{\mathrm{R}} - \rho^{\mathrm{B}})/(\rho^{\mathrm{R}} + \rho^{\mathrm{B}})$ is a color function of fluid interface position, ρ^{B} and ρ^{R} are the densities of blue fluid and red fluid, respectively; σ is the interfacial tension; $\boldsymbol{n}$ is the unit normal vector perpendicular to the fluid interface, $\boldsymbol{n} = -\nabla\rho^N/|\nabla\rho^N|$; κ is the curvature of fluid interface, and its value is related to the unit normal vector $\boldsymbol{n}$, $\kappa = -\nabla_s \cdot \boldsymbol{n}$, ∇_s is the gradient operator of fluid interface.

The total fluid velocity is calculated as:

$$\rho \boldsymbol{u}(\boldsymbol{x},t) = \sum_i f_i(\boldsymbol{x},t)\boldsymbol{c}_i + \frac{1}{2}[\boldsymbol{F}_s(\boldsymbol{x},t) + \boldsymbol{G}]\Delta t \tag{12}$$

To ensure that the immiscible fluids have a reasonable interface with limited thickness, the distribution function of red fluid and blue fluid after recoloring are as follows:

$$f_i^R(x,t) = \frac{\rho_R}{\rho}\tilde{f}_i(x,t) + \beta \frac{\rho_R \rho_B}{\rho} \omega_i \cos(\varphi_i)|c_i| \tag{13}$$

$$f_i^B(x,t) = \frac{\rho_B}{\rho}\tilde{f}_i(x,t) + \beta \frac{\rho_R \rho_B}{\rho} \omega_i \cos(\varphi_i)|c_i| \tag{14}$$

where β is the separation coefficient related to the interface thickness, $0 < \beta \leqslant 1$, the value of β increases with the decrease of interface thickness. To ensure a high accuracy of fluid interface, the value of β equals to 0.7 in this study. φ_i denotes to the angle between discrete velocity and color gradient, rad.

After the propagation step, the distribution function is transferred from the current lattice node to the adjacent lattice node:

$$f_i^k(x + c_i\Delta t, t + \Delta t) = f_i^k(x,t), k = RorB \tag{15}$$

Finally, the fluid density of immiscible two-phase flow is obtained:

$$\rho^k = \sum_i f_i^k, k = RorB \tag{16}$$

In the process of lattice Boltzmann simulation, the treatments of boundary conditions are also very important. Different boundary conditions can significantly affect the accuracy, stability and calculation efficiency (Li et al., 2019; Ning et al., 2019). The commonly used boundary conditions consist of periodic boundary, non-slip boundary, pressure or velocity boundary, wetting boundary, etc. For two-phase flow in porous media, the effect of complex geometry can be well described by imposing the non-slip boundary condition on solid wall. For non-uniform wetting conditions, it is often assumed that two fluids are mixed on the solid wall, and the direction of color gradient at the three-phase contact line is changed to achieve the expected contact angle distribution.

4. Numerical results and discussion

4.1. Model validation

Here, the accuracy of the developed multiple-relaxation-time color-gradient Boltzmann model is validated using three standard benchmarks tests: the Young-Laplace test, the Buckley-Leverett equation, and immiscible two-phase layered flow.

The Young-Laplace test.

The Young Laplace's law represents that when a droplet in gas or a bubble in liquid reaches a steady state, the pressure difference inside and outside of the droplet (bubble) is inversely proportional to the radius R of the droplet (bubble), which can be described by the equation (17). P_i and P_o are the pressures inside and outside of the droplet (bubble), respectively; σ is the interfacial tension coefficient.

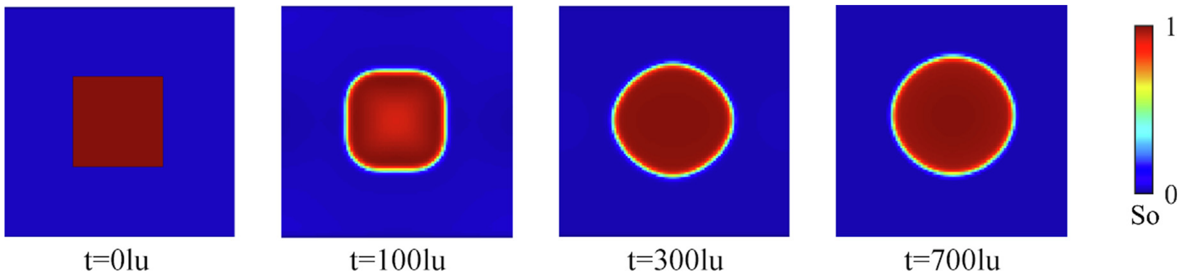


Fig. 3. Morphological evolution of droplets affected by interfacial tension.

$$\Delta p = p_i - p_o = \frac{\sigma}{R} \tag{17}$$

Fig. 3 displays the dynamic evolution of a square droplet under interfacial tension. The calculation area is 100lu × 100lu, the boundary conditions in X and Y directions are set as periodic boundary. The density of oil phase and water phase in lattice units are 1.55 (1550 kg/m^3) and 1.95 (1950 kg/m^3), respectively, and the relaxation time is 0.5. When the LBM simulation reaches a steady state, the oil droplets are stably distributed as a circle. Fig. 4. shows the relationship between the pressure difference Δp and the reciprocal radius $1/R$ of the droplet. The droplet radius is defined as the distance between the point where the density is $(\rho_o + \rho_w)/2$ and the center of the droplet. It demonstrates that the pressure difference Δp and the reciprocal radius $1/R$ satisfy a good linear relationship, indicating that the proposed methodology for two-phase flow can well match the Young Laplace's law. The estimated interfacial tension coefficient is 0.10328 (16.748 mN/m). The lattice units can be converted to physical units based on the scaling amount of the set length, time, and fluid density (Ba et al., 2016).

The Buckley-Leverett equation.

The commonly used Buckley Leverett (B-L) equation represents the immiscible two-phase flow in porous media. Under the same assumptions, the LBM simulation results are compared with the analytical solutions of the B-L equation. In B-L analysis, the residual oil saturation and irreducible water saturation are set to 0.2, and the immiscible displacement is piston-like. Fig. 5. shows the relationship between water saturation and normalized position. It demonstrates that the analytical solutions are in good agreement with the simulated results. There is a shock wave of saturation at the water front in the B-L analytical solutions while the LBM simulated results change continuously along the water front. The analysis indicates that: LBM simulation has a compressibility effect that cannot be ignored, typically leading to a systematic error when estimating water saturation based on two-phase fluid density; Moreover, the numerical diffusion is non-negligible, which can be reduced by introducing a higher-order model or a refined grid system, but it can't be completely eliminated.

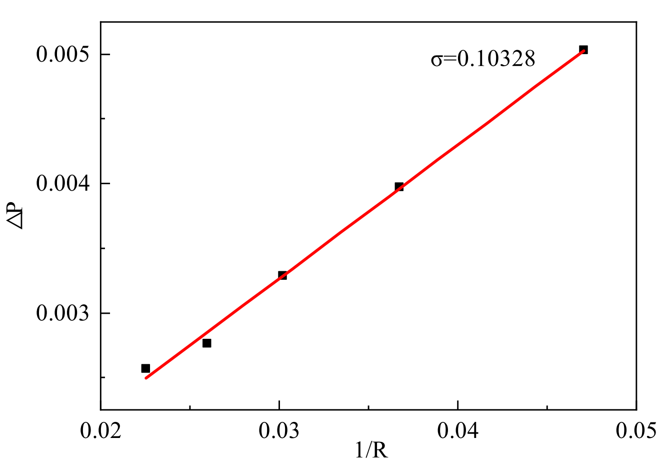


Fig. 4. Relationship between pressure difference Δp and reciprocal radius 1/R of the droplet.

The immiscible two-phase layered flow.

To demonstrate whether the multi-relaxation-time color-gradient Boltzmann model can accurately calculate the fluid velocity distribution, the LBM simulated results are compared with the analytical solutions of immiscible two-phase layered flow. In the study, the non-wetting fluid is distributed in the middle of the calculation domain (100lu × 50lu) while the wetting fluid transports along the boundary of the calculation domain, as shown in Fig. 6. Both the inlet and outlet of the calculation domain is imposed by

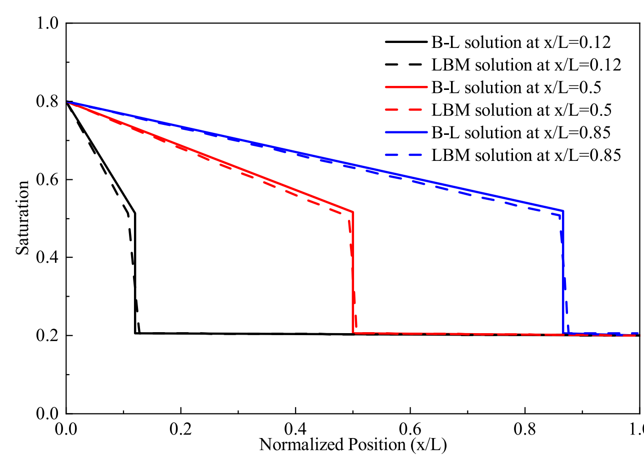


Fig. 5. Comparison between the simulated results and B-L analytical solutions.

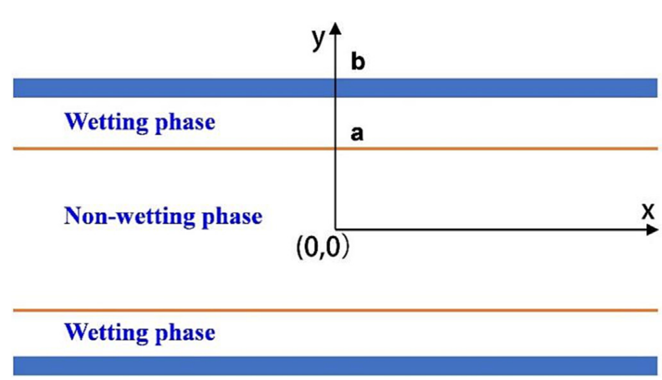


Fig. 6. Immiscible two-phase layered flow.

a periodic boundary, and the solid wall is treated as a semi bounce-back boundary. The fluid flow occurs along the X direction by exerting an external force (F = 1 × 10^{-8}lu). The fluid velocity strongly depends on the viscosity ratio M between the non-wetting fluid and the wetting fluid,$M = \rho_{nw}\upsilon_{nw}/\rho_w\upsilon_w$, where ρ_w and ρ_{nw} are the densities of wetting fluid and non-wetting fluid, respectively; υ_w and υ_{nw} are the dynamic viscosity coefficients of wetting fluid and non-wetting fluid, respectively. both taking a value of 0.16. The analytical expression of fluid velocity is described as follows (Kang et al., 2004; Zou and He, 1997).

$$\begin{aligned} u_{x,nw}(y) &= A_1y^2 + C_1, 0 \leqslant |y| \leqslant a \\ u_{x,w}(y) &= A_2y^2 + B_2y + C_2, a \leqslant |y| \leqslant b \end{aligned} \tag{18}$$

where u_x is the flow velocity along the × direction;$A_1 = -F/2\rho_{nw}\upsilon_{nw}$,$A_2 = -F/2\rho_w\upsilon_w$; $\boldsymbol{F}$ is external force;$B_2 = -2A_2a + 2MA_1a$,$C_1 = (A_2 - A_1)a^2 - B_2(b\ \ - a) - A_2b^2$, $C_2 = -2A_2b^2 - B_2b$.

Fig. 7. shows the velocity distribution at x = 0lu under different viscosity ratios. The solid line denotes to the analytical solution, and the scattered points denote to the LBM simulated results. It can be concluded that, there is a good agreement between the simulated results and the analytical solutions, indicating a strong capacity to predict the fluid velocity distribution during the immiscible two-phase layered flow using the proposed method.

Taking the 3D REV domain (0.9 mm × 0.9 mm × 0.9 mm) retrieved from the digitalized multimodal carbonate rock shown in Fig. 1(b) as the geometry model, the multiple-relaxation-time color-gradient Boltzmann model is used to simulate oil–water two-phase flow in carbonate porous media, and then the effects of rock wettability, oil-wet heterogeneity, capillary number and oil–water viscosity ratio on pore-scale oil displacement efficiency and fluid distribution are discussed. There exist 300 lattice grids along X, Y, and Z directions with a grid size of 3.0 μm. In the process of LBM simulation, initial oil saturation is 0.85, irreducible water saturation is 0.15, oil phase viscosity is 5.0 mPa s, and water-phase viscosity is 1.0 mPa s. The densities of two-phase fluids are the same, both of which are 1000 kg/m^3. The rock walls are strongly oil-wet with a certain value of contact angle. For the basic case, the contact angle is achieved as 157.5° by exerting the wetting boundary condition. The water front migrates downward along the z-axis. By imposing the Zou-He (Liu et al., 2017) boundary condition, it is ensured that any X-Y cross section has a constant volume flux. The capillary number is 1.0 × 10^{-4}. Periodic boundary is exerted at the X and Y directions, and non-slip boundary is imposed on rock solid walls.

4.2. Effect of rock wettability

In this study, strong oil-wet (θ = 157.5°), intermediate-wet (θ = 90°) and strong water-wet (θ = 22.5°) are considered. The other properties and boundary conditions remain unchanged. The multiple-relaxation-time color-gradient Boltzmann model is used to simulate oil–water two-phase flow, and the influence of rock wettability on pore-scale oil displacement efficiency and fluid distribution in carbonate porous media is analyzed. The dynamic variations of pore-scale oil displacement efficiency under different wetting conditions are shown in Fig. 8. The two-phase fluid distributions are displayed in Fig. 9. The results demonstrate that, the retained oil content in pore space of the carbonate rock decreases gradually as water injection proceeds; The pore-scale oil displacement efficiency is higher as the rock walls become more water-wet. For the water-wet case, capillary pressure act as the driving force for oil displacement, which cooperates with viscous force to effectively mobilize and recover the oil droplets. As the injected water preferentially moves downward the rock walls, oil droplets occupied in pore space are gradually stripped, and residual oil droplets are mainly distributed in the middle of pores and throats. For the oil-wet case, due to that capillary pressure serves as the resistance of flow, oil droplets in pore space can be recovered only if the capillary pressure is overcome and the residual oil droplets are mainly distributed in the local locations contacting with the rock wall. For the intermediate-wet case, capillary pressure equals to zero, so viscous force is the only driving force to recover oil droplets in pore space. Therefore, the oil displacement efficiency falls in between oil-wet and water-wet cases. The above research shows that wettability is not the only factor restricting oil displacement efficiency, the complexity of reservoir microscopic pore structure has a strong impact on oil displacement efficiency.

To further reveal the underlying mechanism of rock wettability on pore-scale oil droplet mobilization in pore space, the volume

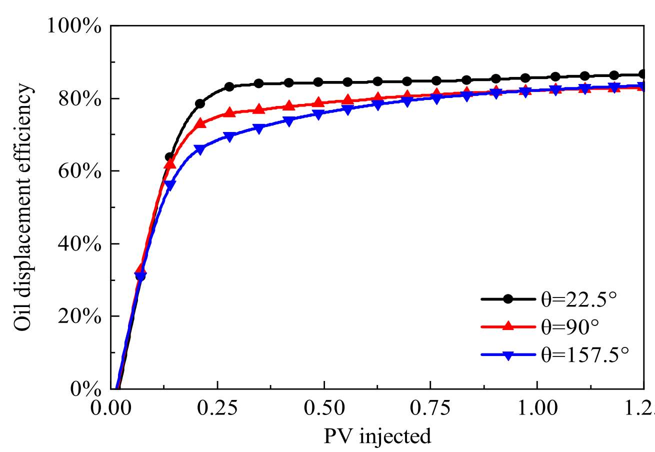
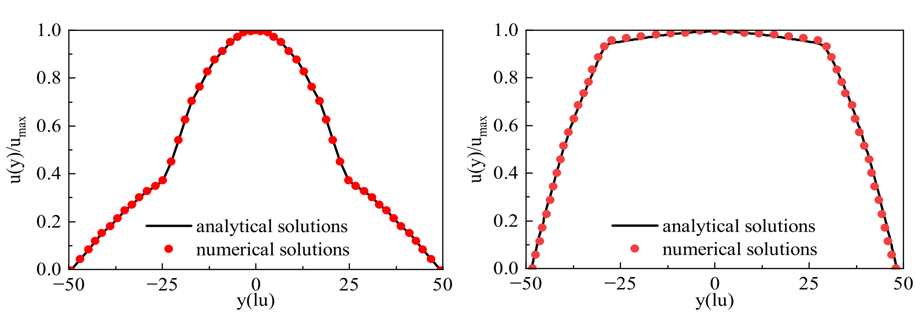


Fig. 8. Oil displacement efficiency in carbonate rock under different wetting conditions.



(a) M=0.2 **(b) M=50**

Fig. 7. Velocity distribution of immiscible two-phase layered flow at x = 0 lu.

and Euler coefficient (Rabbani et al., 2018) of continuous and isolated oil droplets are calculated by the integral geometry, the formula of Euler coefficient is as follows:

$$\chi = \frac{M_3}{4\pi} = \beta_0 - \beta_1 + \beta_2 \tag{19}$$

Where χ is the Euler coefficient; M_3 is the integral Gaussian surface; β_0 is the number of isolated oil droplets in the REV region; β_1 is the number of pore throats associated with the network-like oil droplets; β_2 is the number of isolated water droplets surrounded by injected water. The negative Euler coefficient represents that the microstructure of oil has strong topological connectivity, implying a network-like oil droplet. The positive Euler coefficient imply the presence of disconnected oil droplets, implying an isolated oil droplet.

The calculation results are shown in Fig. 10 and Fig. 11. It demonstrates that, compared with water-wet and intermediate-wet cases, the volume of continuous oil droplets in oil-wet core decreases more slowly, the topology of continuous oil droplets in oil-wet core is not easily manipulated since it wets the grain surfaces and remains connected. Therefore, under the oil-wet condition, the starting point of the Euler coefficient of the continuous oil droplets is low. At the later stage of water injection, the Euler coefficient of the isolated oil droplets has a significant downward trend, and the droplets return to the connected state. The volume of isolated oil droplets increased in the early stages and then decreased, when residual oil saturation is achieved, the volume of continuous oil droplets is basically the same under different wetting conditions, while the volume of isolated oil droplets in strong water-wet core is much smaller. As displayed in Fig. 10 (b), the peak value of the isolated oil droplets in the water-wet core is low. A large number of continuous oil droplets are not broken and move out with the water injection, the volume of the isolated oil droplets increases less in a short time, which is also the reason for the high oil displacement efficiency.As displayed in Fig. 11, the Euler coefficient of continuous oil droplets in pore space increases

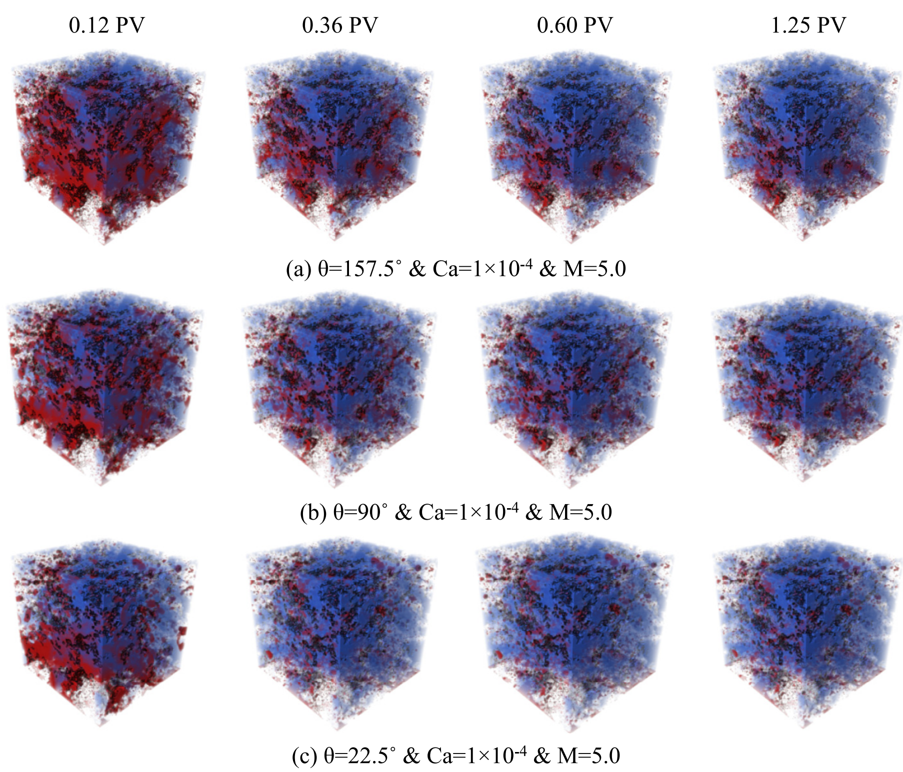


Fig. 9. Effect of rock wettability on two-phase fluid distribution in carbonate pore space.

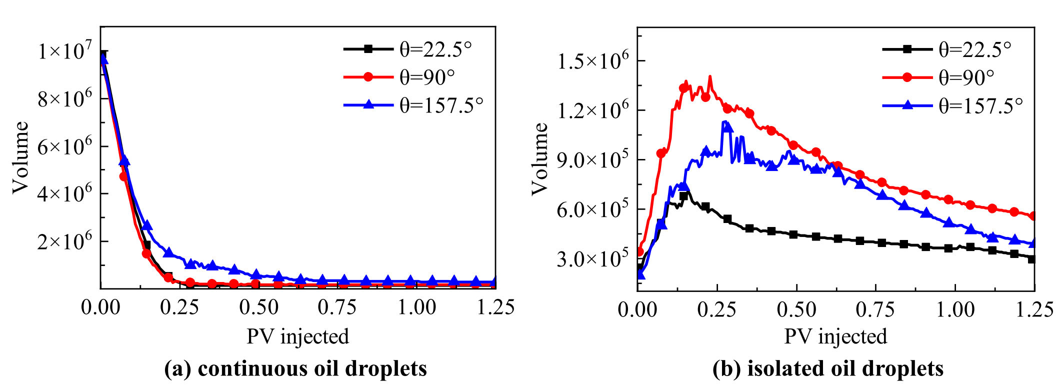


Fig. 10. Volume variation of residual oil droplets under different wetting conditions.

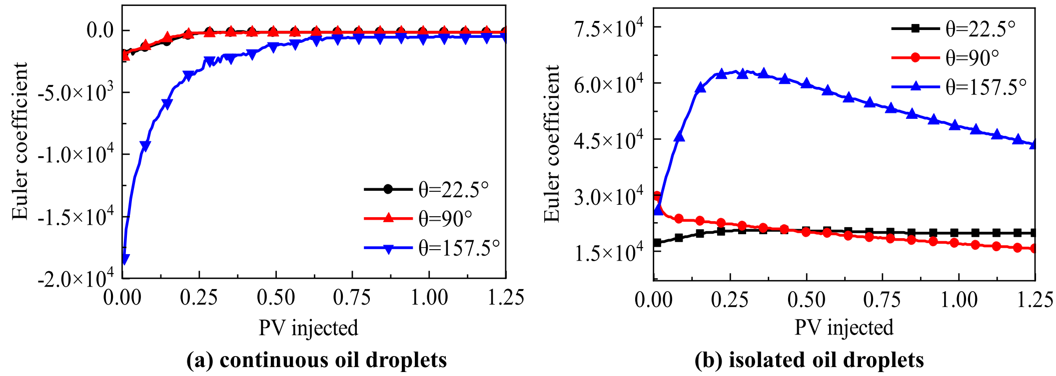


Fig. 11. Euler coefficient variation of oil droplets under different wetting conditions.

gradually as the pore volume of injected water magnifies, meaning that the topological connectivity of oil droplets becomes worse. In contrast with strong water-wet and intermediate-wet cores, the strong oil-wet core typically shows a smaller Euler coefficient of continuous oil droplets in pore space under the same PV injected, indicating that more continuous oil droplets with strong topological connectivity are still waiting for strip-out. In intermediate-wet core, the topology is more easily manipulated by viscous forces. Therefore, the Euler coefficients of strong water-wet core and intermediate-wet core have little change, and the oil droplets remain isolated. In the previous study of Rabbani et al. (Yang et al., 2021), pore geometry plays a crucial role in governing the apparent wettability under intermediate-wet conditions, which leads to co-existence of fluid interfaces with different fluid shapes. Our research shows that the change trend of isolated oil droplets connectivity in intermediate-wet core is different from that of water-wet core and oil-wet core.

4.3. Effect of oil-wet heterogeneity

Three oil-wet heterogeneity cases are designed with the help of a normal distribution function (as shown in Fig. 12). The average contact angles are respectively 157.5°, 137.5° and 117.5° with a same variance of 2.25. The other properties and boundary conditions used for LBM simulation are consistent with the basic case described above. To investigate the effect of oil-wet heterogeneity on pore-scale oil displacement efficiency and fluid distribution in carbonate porous media, the multiple-relaxation-time color gradient Boltzmann model is used to simulate oil–water two-phase flow

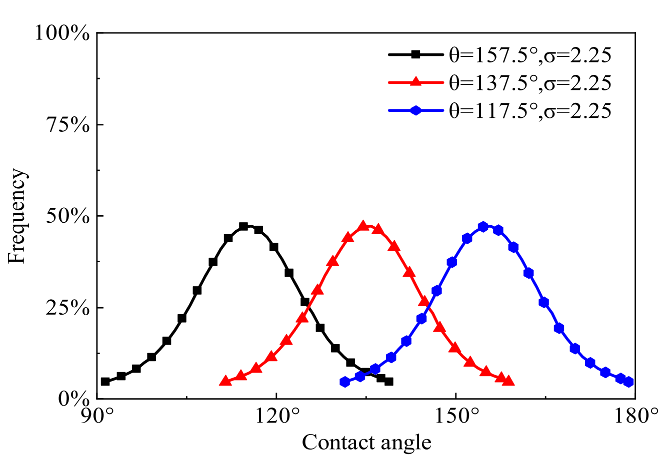


Fig. 12. Distribution of oil-wet heterogeneity in carbonate porous media.

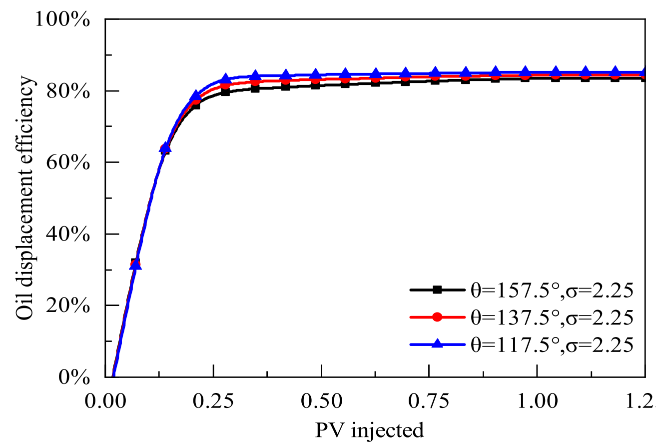


Fig. 13. Effect of oil-wet heterogeneity on pore-scale oil displacement efficiency.

in the 3-D REV domain. Fig. 13 plots the variations of pore-scale oil displacement efficiency under the influence of oil-wet heterogeneity, and the two-phase fluid distribution in carbonate porous media is shown in Fig. 14.

It is not difficult to observe that the oil-wet heterogeneity only has a slight impact on the pore-scale oil displacement efficiency in the multimodal carbonate rock. The degree of oil-wetting becomes higher as the average contact angle increases, resulting in a lower oil displacement efficiency under the same PV injected. For the oil-wet carbonate rock samples, the capillary force serves as a resistance of flow to inhibit the recovery of oil droplets at pore-scale during waterflooding; The capillary force is gradually raised with the increase of contact angle, leading to a reduction of effective driving force of crude oil under the same capillary number; The oil content stripped out of pore space decreases while the residual oil content retained in pore space gradually increases. However, the dynamic evolution of oil–water two-phase fluid distribution is basically consistent in the process of waterflooding.

Fig. 15 and Fig. 16 reflect the influence of oil-wet heterogeneity on dynamic variation of the volume and Euler coefficient of oil droplets occupied in carbonate pore space. For different oil-wet heterogeneity conditions, the volume and Euler coefficient of continuous and isolated oil droplets in pore space exhibit a similar change trend. In the process of water flooding, the volume of continuous oil droplets gradually decreases and tends to be stable, while the volume of isolated oil droplets increases at the early stages and then decreases. At the initial stage of water injection, a large number of oil droplets with good connectivity are dispersed, and the Euler coefficient becomes positive. After 0.25 PV,

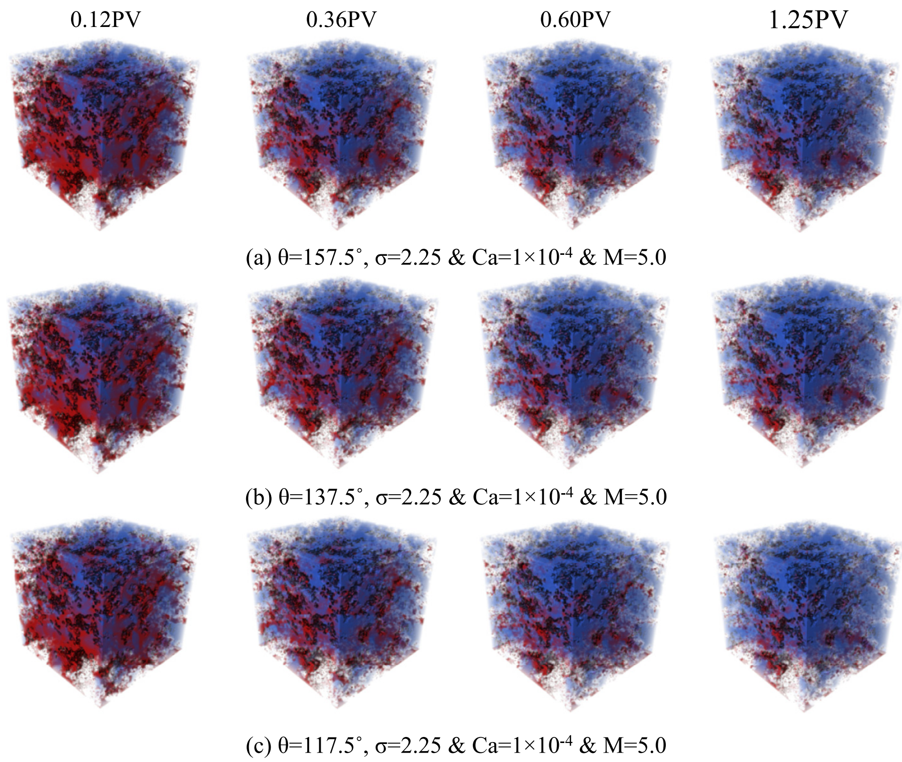


Fig. 14. Effect of oil-wet heterogeneity on two-phase fluid distribution in carbonate pore space.

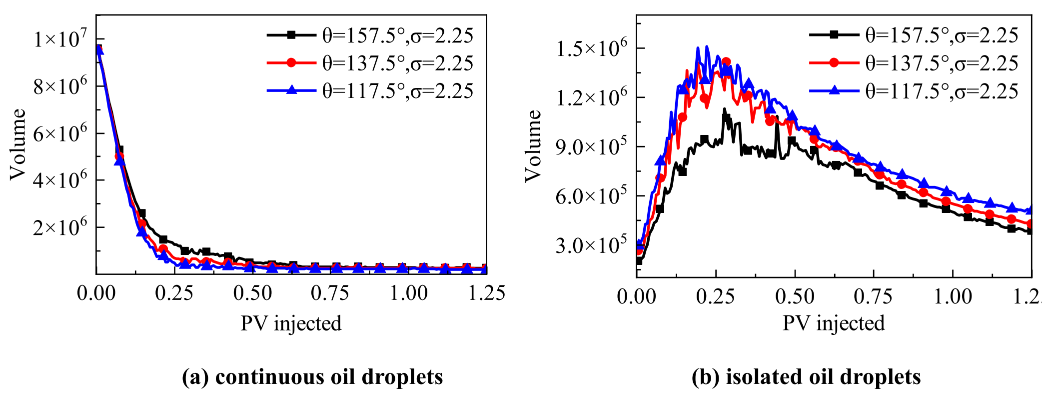


Fig. 15. Variation in volume of oil droplets under different oil-wet heterogeneity.

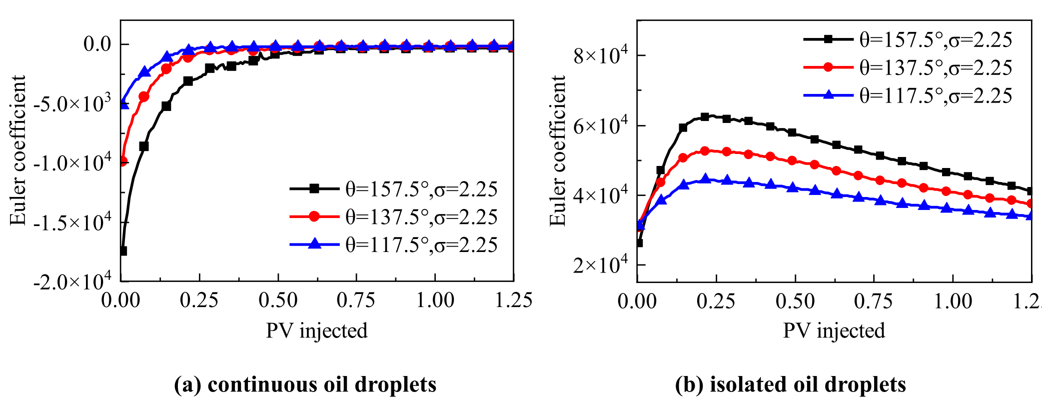


Fig. 16. Variation in Euler coefficient of oil droplets under different oil-wet heterogeneity.

some isolated oil drops are washed away by water injection, and the connectivity is enhanced. Strong oil-wet oil droplets still remain on the rock surface in the form of poor connectivity. The Euler coefficient of continuous oil droplets sharply increases and then tends to be a constant value. The change in the Euler coefficient of isolated oil droplets is consistent with that of isolated oil droplets' volume. Under the same PV of injected water, the volume of continuous oil droplets is larger and the corresponding topological connectivity is better as the rock walls become more oil-wet while the topological connectivity of isolated oil droplets becomes worse.

4.4. Effect of capillary number

To explore the influence of capillary number on the pore-scale oil displacement efficiency and fluid distribution in carbonate porous media, three cases with different capillary numbers is numerically simulated using the proposed multiple-relaxation-time color-gradient LB model. For this study, the capillary numbers equal to 1×10^{-5}, 5×10^{-5} and 1×10^{-4}, respectively. The other properties and boundary conditions used for LBM simulation are consistent with the basic case described above. The dynamic variations of pore-scale oil displacement efficiency under the influence of different capillary numbers are shown in Fig. 17. The corresponding two-phase fluid distributions in carbonate pore space are displayed in Fig. 18.

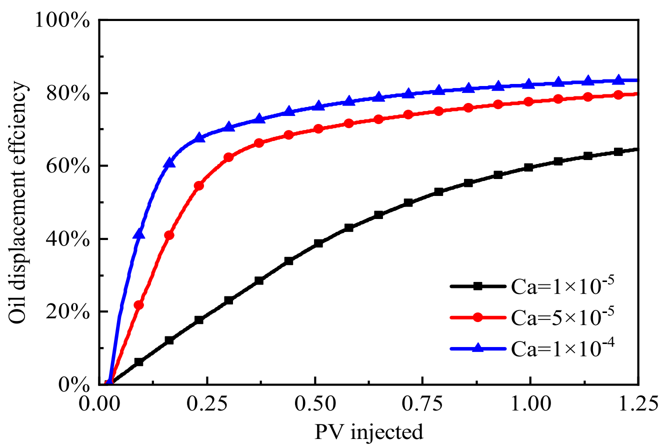


Fig. 17. Effect of capillary number on pore-scale oil displacement efficiency.

As displayed in Fig. 17 and Fig. 18, the capillary number has a significant impact on pore-scale oil displacement efficiency of carbonate rocks, and the two-phase fluid distribution strongly depends on the dynamic competition between the capillary pressure and the viscous force at pore scale. The larger the volume flux of the displacing fluid assigned downward along the Z direction is, the higher the capillary number is and a higher effective driving force can be achieved. It makes the oil droplets occupied in carbonate pore space much easier to be stripped and recovered, implying a better pore-scale oil displacement efficiency. When the capillary number equals to 5×10^{-5} and 1×10^{-4}, the majority of oil droplets occupied in oil-wet carbonate rocks are recovered at the early stage of water injection. Once the cumulative PV of injected water is higher than 1.25 PV, the two-phase flow will reach a steady state. The change of capillary number is realized by adjusting the injection rate. Higher capillary number means higher flow velocity in pore space and shorter displacement time. When the capillary number equals to 1×10^{-5}, because of the competition between the capillary pressure and the viscous force, merely about 60 % of crude oil are recovered at 1.25PV, and the two-phase flow has not yet reached a steady state, indicating that a large quantity of oil droplets retained in pore space of the multimodal carbonate rock. Similar results were observed in the study of Yang et al. [57].

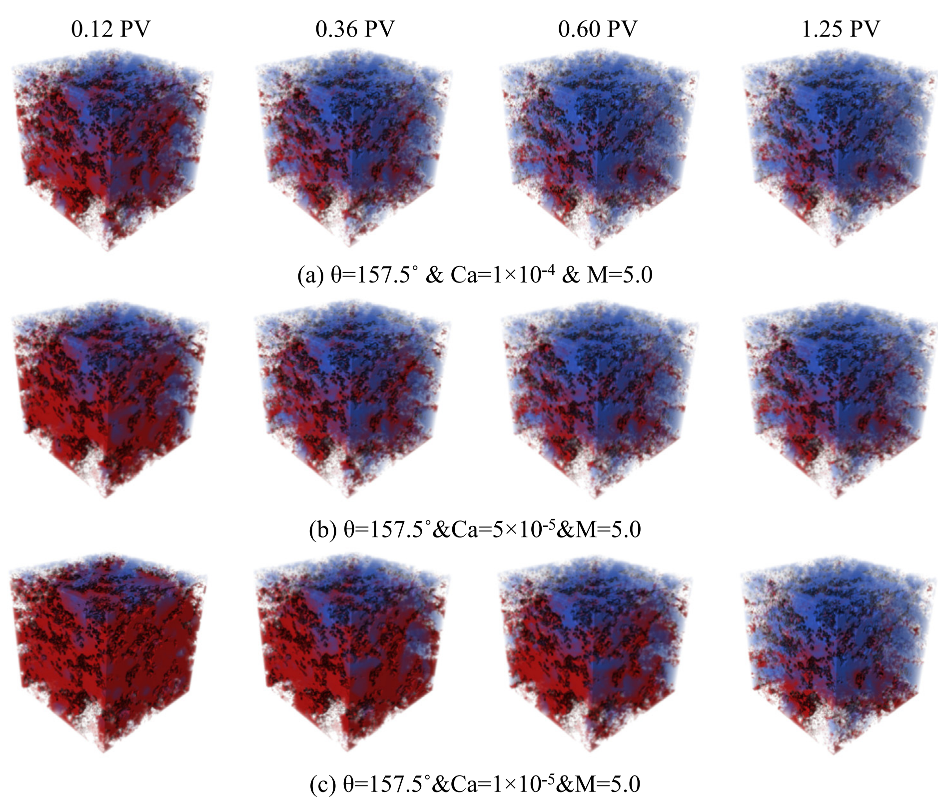


Fig. 18. Effect of capillary number on two-phase fluid distribution in carbonate pore space.

Fig. 19 and Fig. 20 depict the dynamic variation in volume and Euler coefficient of pore-scale oil droplets under the influence of capillary number, respectively. If the capillary number takes the value of 5×10^{-5} or 1×10^{-4}, the volume and Euler coefficient of continuous and isolated oil droplets occupied in pore space show a similar variation trend. At the early stages of displacement, the continuously distributed oil droplets are fragmented and broken into many small isolated oil droplets, resulting in a rapid reduction of the volume of continuous oil droplets, a sharp increase of the volume of isolated oil droplets and a worse topological connectivity of single oil droplets. As water injection continues, more isolated oil droplets are stripped and recovered by the injected water, leading to a smaller volume and a stronger topological connectivity of isolated oil droplets. For the case with capillary number of 1×10^{-5}, the dynamic competition between oil and water makes the volume and Euler coefficient of oil droplets change slowly, aggravating the difficulty to mobilize pore-scale oil droplets.

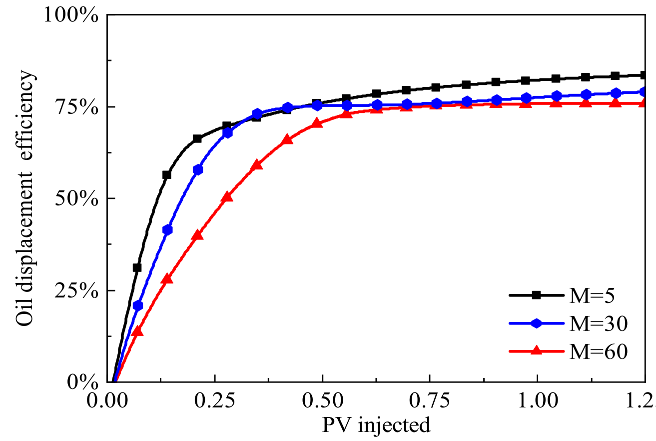
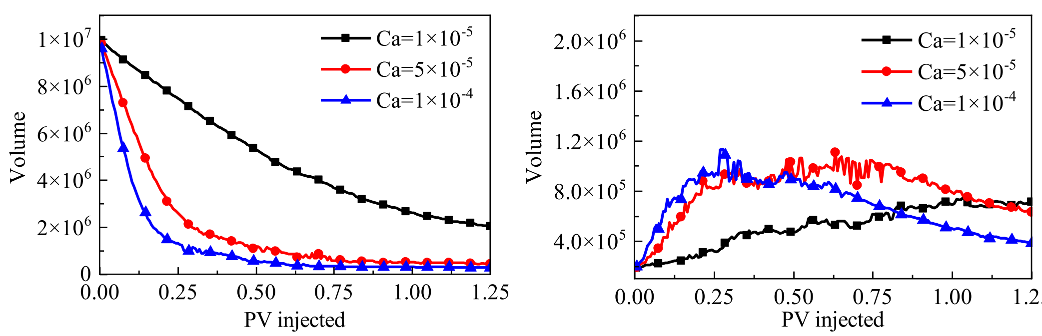


Fig. 21. Effect of oil–water viscosity ratio on pore-scale oil displacement efficiency.

4.5. Effect of oil–water viscosity ratio

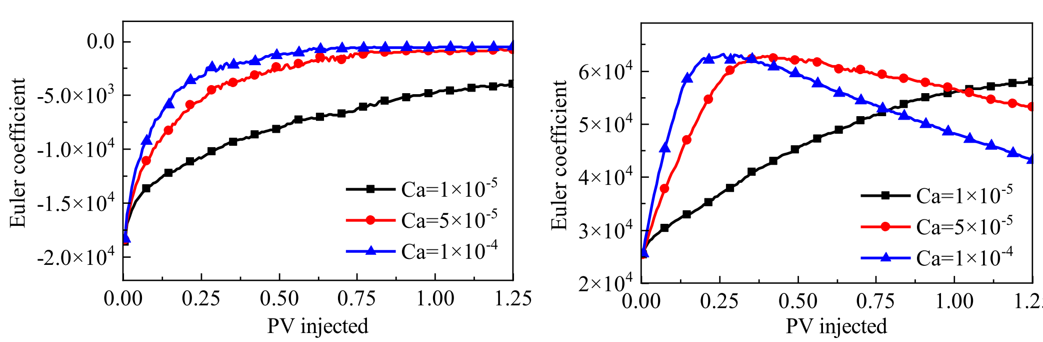
To explore the influence of oil–water viscosity ratio on pore-scale oil displacement efficiency and fluid distribution in carbonate porous media, The proposed multiple-relaxation-time color-gradient LB model is used to simulate oil–water two-phase flow under three cases of oil–water viscosity ratios equaling to 5.0, 30.0 and 60.0, respectively. The other properties and boundary conditions in the process of LBM simulation are consistent with the basic case described above. The dynamic variations of pore-scale oil displacement efficiency under the influence of different oil–water viscosity ratios are shown in Fig. 21. The corresponding two-phase fluid distributions in pore space of the carbonate rock are displayed in Fig. 22.

As can be seen from Fig. 21 and Fig. 22, for the oil-wet carbonate core, the higher the oil–water viscosity ratio, the lower the pore-scale oil displacement efficiency under the same PV of injected water. This is mainly because that with the increase of oil–water viscosity ratio, the flow capability of oil droplets occupied in carbonate pore space becomes worse, and the difference in mobility of water and oil tends to be larger, which significantly aggravates the difficulty to mobilize the oil droplets. Moreover, once the



(a) continuous oil droplets **(b) isolated oil droplets**

Fig. 19. Variation in volume of oil droplets under different capillary numbers.



(a) continuous oil droplets **(b) isolated oil droplets**

Fig. 20. Variation in Euler coefficient of oil droplets under different capillary numbers.

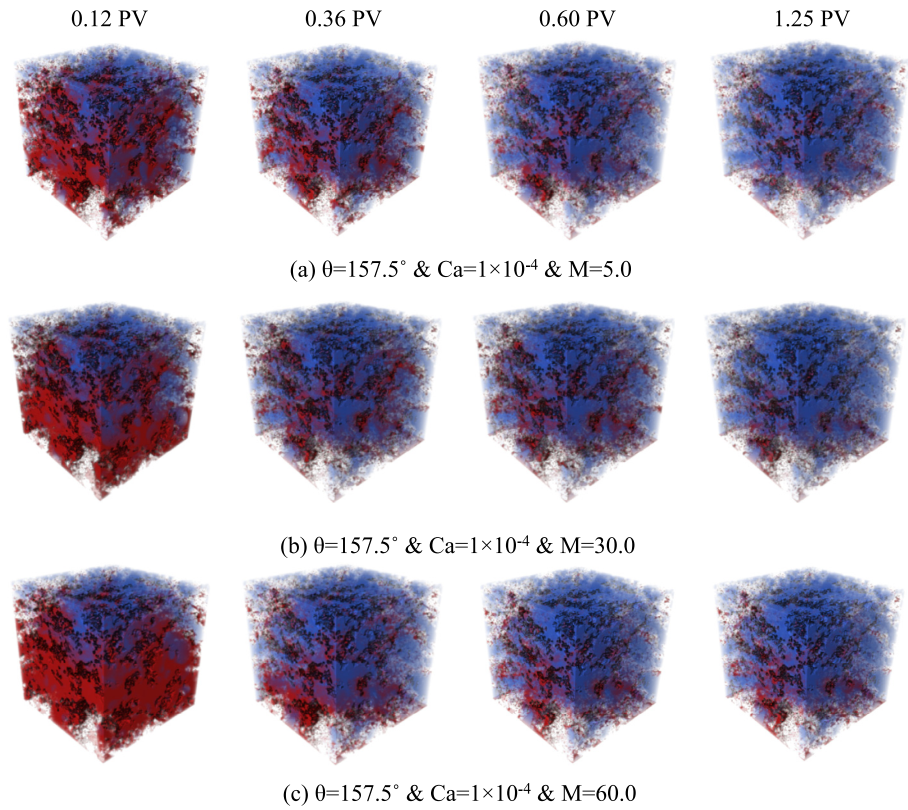


Fig. 22. Effect of oil–water viscosity ratio on fluid distribution in carbonate pore space.

injected water forms a preferential flow pathway in pore space, ineffective water circulation occurs and only a limited sweep zone of oil droplets at pore-scale is achieved. It implies that the oil–water viscosity ratio can exert an important impact on the pore-scale displacement efficiency of the carbonate rock. According to Fig. 22 (a), when the viscosity ratio of oil and water is small, multiple paths are formed from the inlet to the outlet, resulting in a higher oil displacement efficiency. The larger the oil–water viscosity ratio, the stronger the dynamic inhomogeneity as the displacement process proceeds.

Fig. 23 and Fig. 24 reflect the influence of oil–water viscosity ratio on the volume and the Euler coefficient of oil droplets in pore space of the carbonate rock, respectively. For the cases with different oil–water viscosity ratios, the volume and Euler coefficient of continuous and isolated oil droplets in pore space of the carbonate rock have the same change trend. With the continuous increase of the injected pore volume, large oil droplets that are mainly distributed in neighboring pores and throats are gradually broken into many isolated oil droplets, resulting in a rapid reduction of the volume of continuous oil droplets and a sharp deterioration of topological connectivity. However, the dynamic change in the volume and Euler coefficient of isolated oil droplets occupied in pore space of the carbonate rock can be subdivided into two stages: at the early stage of water injection, the volume of isolated oil droplets increases sharply followed by a worse topological connectivity. As the pore volume of injected water into the carbonate rock increases, the isolated oil droplets are gradually stripped out, leading to a remarkable reduction of the volume and improvement of the topological connectivity. When the oil–water viscosity ratio is 5, the volume of the continuous oil droplets decreases rapidly, while the Euler coefficient is similar to that when the oil–water viscosity ratio is 30 and 60. It can be seen that the higher oil dis-

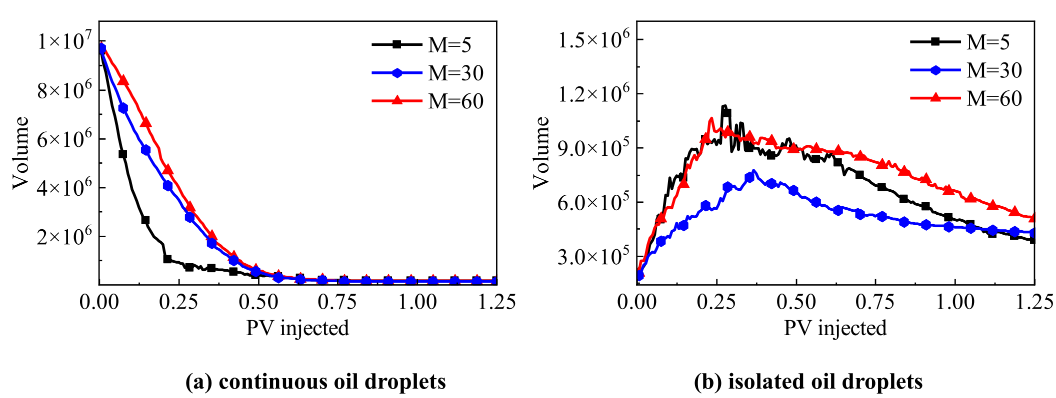
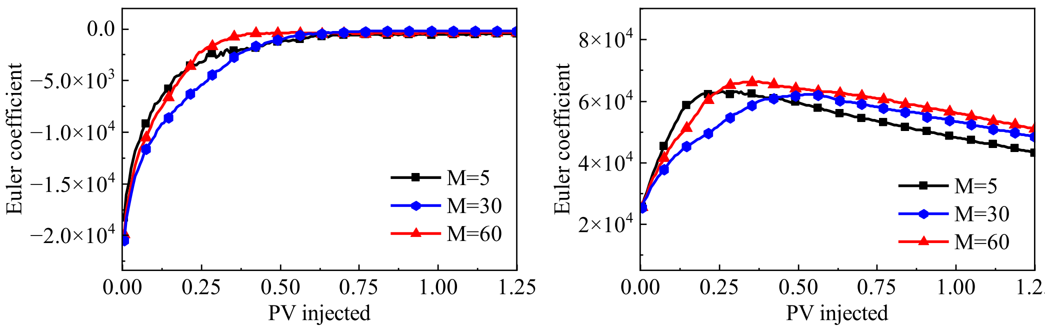


Fig. 23. Variation in volume of oil droplets under different oil–water viscosity ratios.



(a) continuous oil droplets (b) isolated oil droplets

Fig. 24. Variation in Euler coefficient of oil droplets under different oil–water viscosity ratios.

placement efficiency is mainly due to more fluid paths, not changes in fluid structure.

5. Conclusions

In this study, a multiple-relaxation-time color-gradient Boltzmann model is developed to simulate oil–water two-phase flow in multimodal carbonate porous media. The effects of rock wettability, oil-wet heterogeneity, capillary number and oil–water viscosity ratio on pore-scale oil displacement efficiency and fluid distribution are explored, following by analysis of pore-scale oil droplet mobilization by integral geometry. The main conclusions are as follows:

With the increase of injected pore volumes, oil droplets occupied in pore space of the carbonate rock are gradually stripped out and recovered, indicating a gradual improvement of pore-scale oil displacement efficiency. Compared with water-wet and intermediate-wet cases, the volume of continuous oil droplets in oil-wet core decreases more slowly, the topology of continuous oil droplets in oil-wet core is not easily manipulated.

The rock wettability, capillary number and oil–water viscosity ratio have significant impacts on the pore-scale oil displacement efficiency and fluid distribution in multimodal carbonate pore space while oil-wet heterogeneity has little effect. The oil displacement efficiency usually becomes larger with the increase of capillary number and the water-wetting degree as well as the decrease of oil–water viscosity ratio. For water-wet cores, the residual oil droplets are mainly distributed in the middle of pores and throats; for oil-wet cores, the residual oil droplets mainly occur in the positions contacting with the rock walls.

Due to dynamic competition between capillary pressure and viscous force, the continuous oil droplets occupied in the carbonate pore space are fragmented into a large quantity of isolated oil droplets in the early stage of water flooding, showing a sharp decrease in the volume of continuous oil droplets, a rapid increase in the volume of isolated oil droplets, and a poor topological connectivity; In the middle and late stage of water flooding, the isolated oil droplets are gradually stripped and mobilized, leading to a decrease in the volume of oil droplets and an obvious improvement in topological connectivity.

CRediT authorship contribution statement

Daigang Wang: Conceptualization, Methodology, Writing – original draft. **Fangzhou Liu:** Software, Validation, Writing – original draft. **Jingjing Sun:** Investigation, Data curation, Writing – review & editing. **Yong Li:** Resources, Supervision, Funding acquisition. **Qi Wang:** Validation. **Yuwei Jiao:** Visualization. **Kaoping Song:** Supervision, Funding acquisition. **Shu Wang:** Investigation. **Ruicheng Ma:** Formal analysis.

Data availability

Data will be made available on request.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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